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FACULTAD DE INGENIERÍA
INGENIERÍA EN SISTEMAS
LABORATORIO REDES DE COMPUTADORAS 2
SEGUNDO SEMESTRE DE 2026



Tarea 3

Introducción a redes inalámbricas y redundancia de capa 3

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Topología

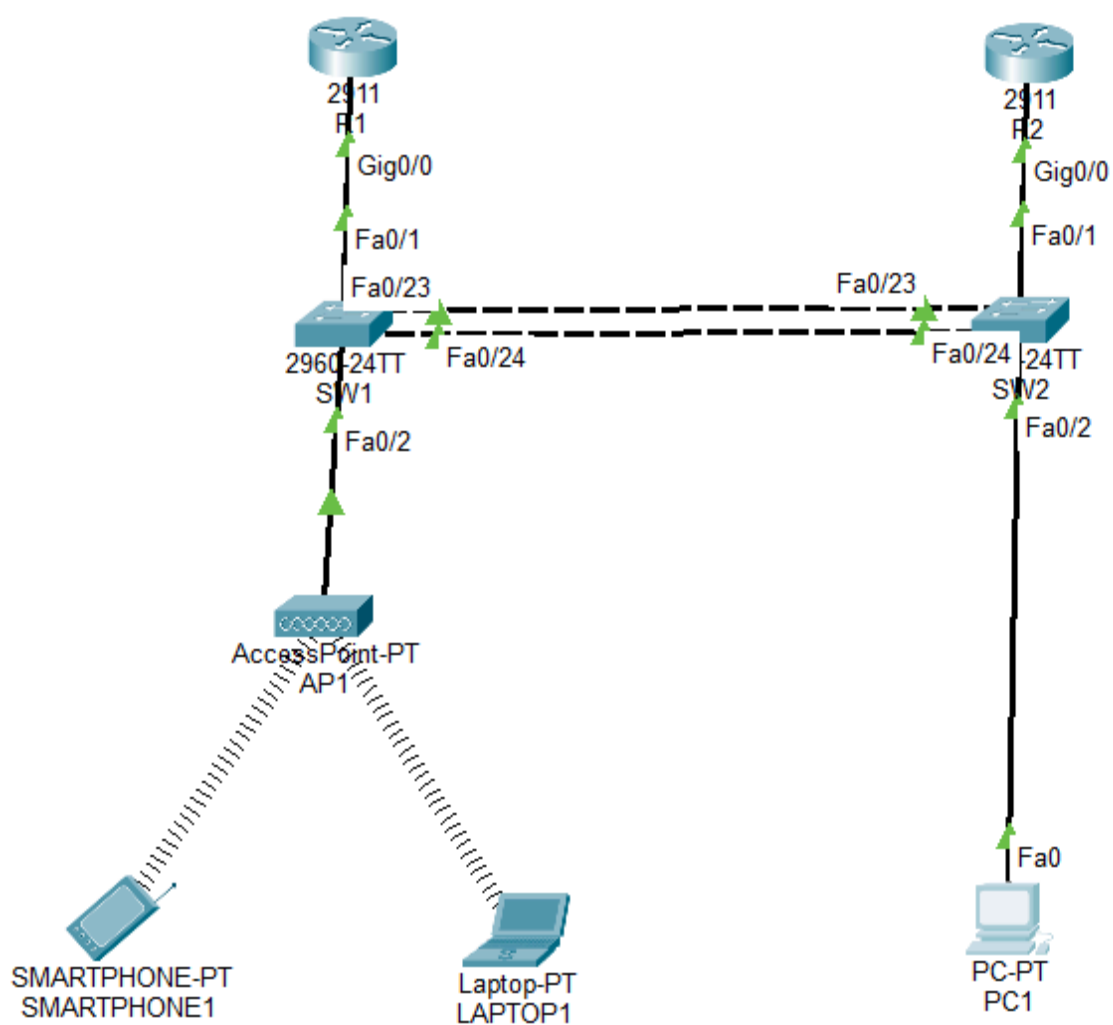


Tabla de direccionamiento IP

Dispositivo	Interfaz	Dirección IP	Máscara	Gateway
R1	Gi0/0	192.168.10.2	255.255.255.0	n/a
R2	Gi0/0	192.168.10.3	255.255.255.0	n/a
IP virtual HSRP	n/a	192.168.10.1	255.255.255.0	n/a
SW1	VLAN 1 (SVI)	192.168.10.11	255.255.255.0	192.168.10.1
SW2	VLAN 1 (SVI)	192.168.10.12	255.255.255.0	192.168.10.1
AP1	n/a	Sin IP (capa 2)	n/a	n/a
PC1	Fa0	192.168.10.10	255.255.255.0	192.168.10.1
LAPTOP1	Wireless0	192.168.10.20	255.255.255.0	192.168.10.1
SMARTPHONE1	Wireless0	192.168.10.21	255.255.255.0	192.168.10.1

Parámetros de HSRP

Parámetro	R1	R2
Grupo HSRP	1	1
IP virtual	192.168.10.1	192.168.10.1
Prioridad	110	100 (por defecto)
Preempt	Sí	No es necesario
Rol esperado	Activo	Standby

Parámetros de EtherChannel y Wi-Fi

Elemento	Valor
Enlace agrupado	SW1 Fa0/23-24 ⇌ SW2 Fa0/23-24
Port-Channel	1
Protocolo	LACP (modo active en ambos switches)
SSID	Redes2_T3_201504070
Seguridad	WPA2-PSK (mínimo 8 caracteres)

Configuración

SW1

```
enable
configure terminal
hostname SW1
interface range fastEthernet 0/23 - 24
  channel-group 1 mode active
exit
interface vlan 1
  ip address 192.168.10.11 255.255.255.0
  no shutdown
exit
end
copy running-config startup-config
write memory
exit
```

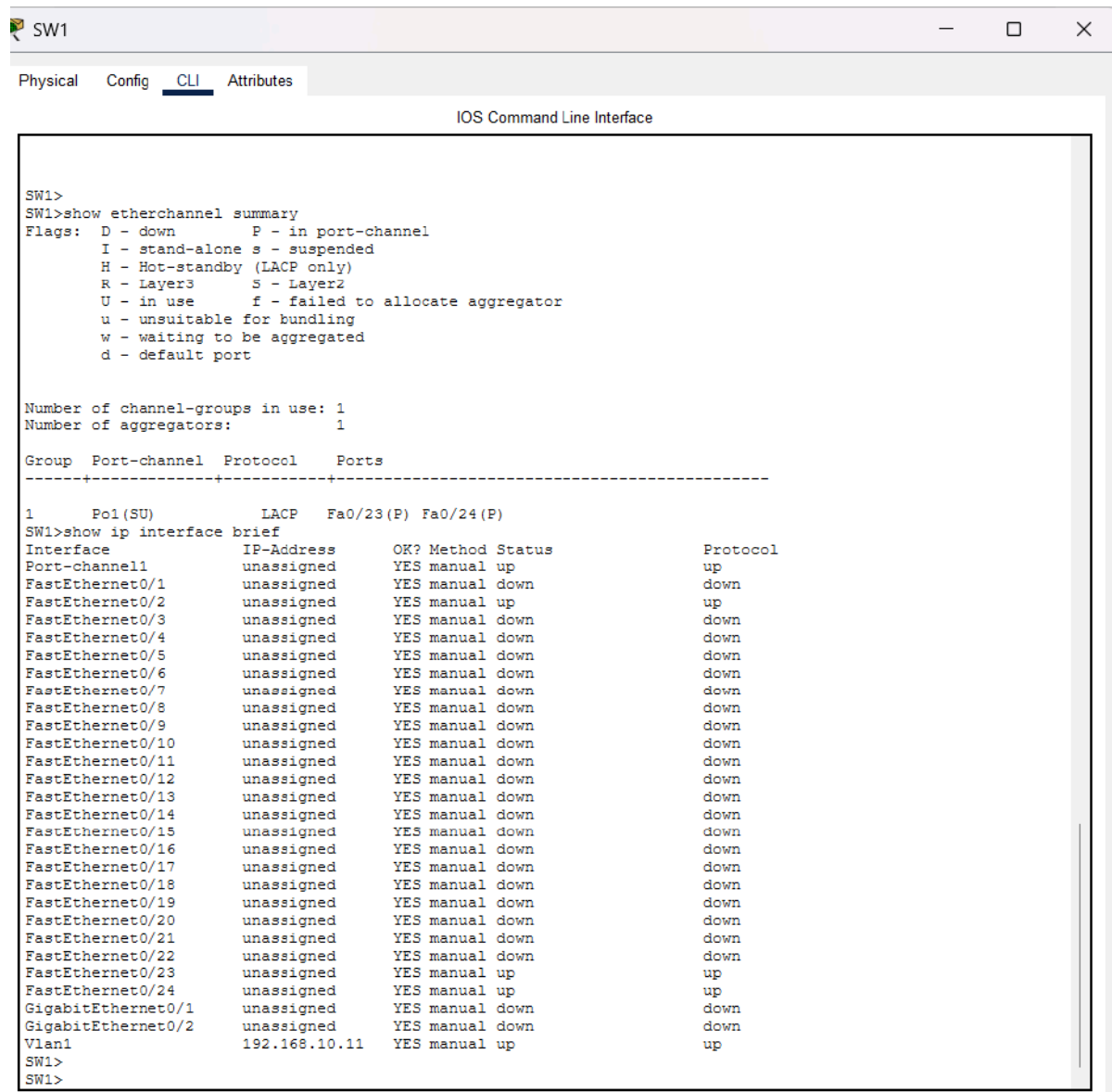
SW2

```
enable
configure terminal
hostname SW2
interface range fastEthernet 0/23 - 24
  channel-group 1 mode active
exit
interface vlan 1
  ip address 192.168.10.12 255.255.255.0
  no shutdown
```

```
exit
end
copy running-config startup-config
write memory
exit
```

Verificación en SW1

```
show etherchannel summary
show ip interface brief
```



The screenshot shows a network switch (SW1) CLI interface. The top bar indicates the switch is in the 'CLI' tab. The main window displays the output of two commands: 'show etherchannel summary' and 'show ip interface brief'.

show etherchannel summary output:

```
SW1>show etherchannel summary
Flags: D - down        P - in port-channel
       I - stand-alone  S - suspended
       H - Hot-standby (LACP only)
       R - Layer3       S - Layer2
       U - in use       f - failed to allocate aggregator
       u - unsuitable for bundling
       w - waiting to be aggregated
       d - default port

Number of channel-groups in use: 1
Number of aggregators:          1

Group  Port-channel  Protocol    Ports
-----+-----+-----+-----
1      Po1(SU)        LACP        Fa0/23(P) Fa0/24(P)
```

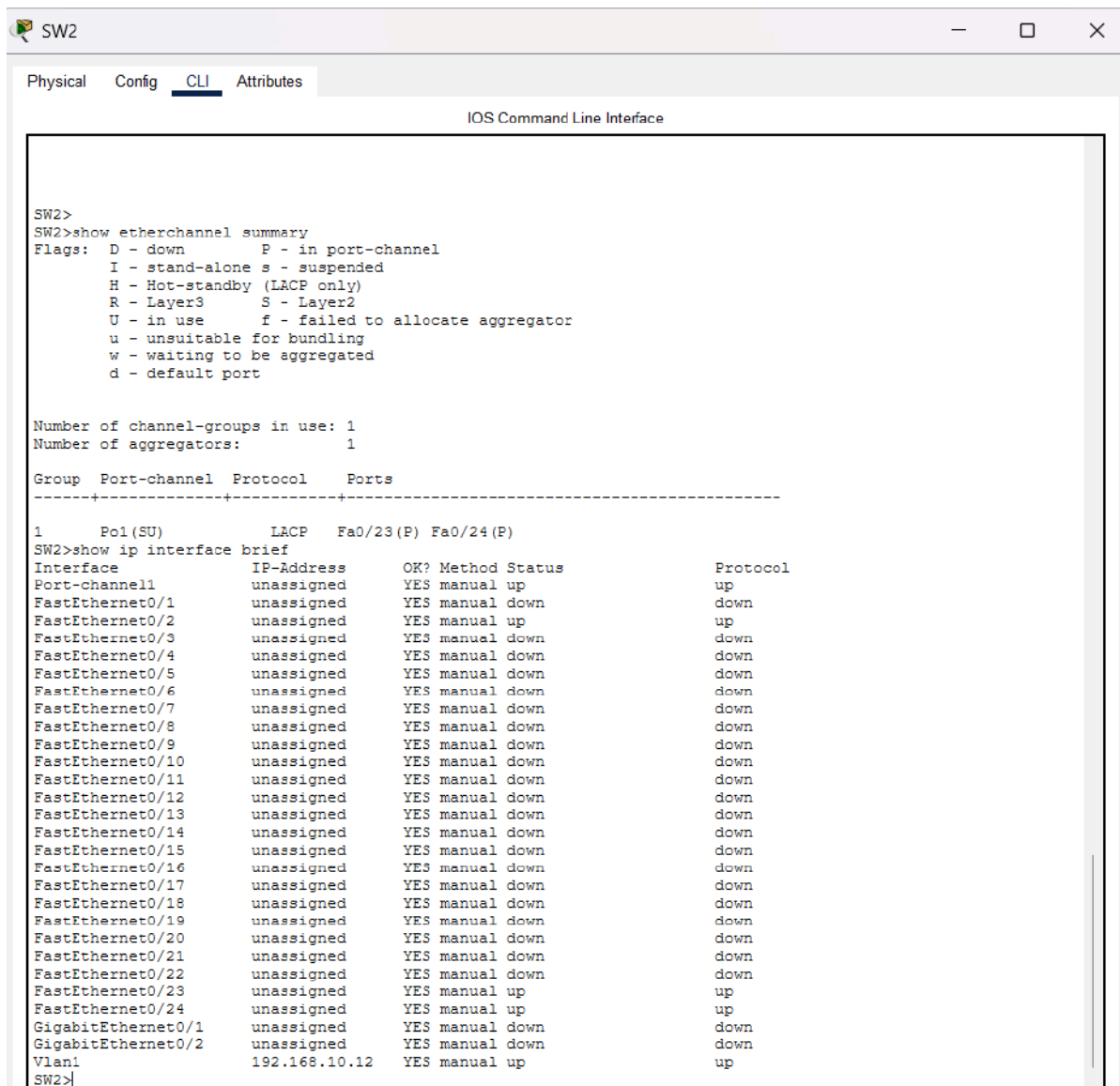
show ip interface brief output:

```
SW1>show ip interface brief
Interface      IP-Address      OK? Method Status  Protocol
Port-channel1  unassigned      YES manual  up      up
FastEthernet0/1 unassigned      YES manual  down    down
FastEthernet0/2 unassigned      YES manual  up      up
FastEthernet0/3 unassigned      YES manual  down    down
FastEthernet0/4 unassigned      YES manual  down    down
FastEthernet0/5 unassigned      YES manual  down    down
FastEthernet0/6 unassigned      YES manual  down    down
FastEthernet0/7 unassigned      YES manual  down    down
FastEthernet0/8 unassigned      YES manual  down    down
FastEthernet0/9 unassigned      YES manual  down    down
FastEthernet0/10 unassigned      YES manual  down    down
FastEthernet0/11 unassigned      YES manual  down    down
FastEthernet0/12 unassigned      YES manual  down    down
FastEthernet0/13 unassigned      YES manual  down    down
FastEthernet0/14 unassigned      YES manual  down    down
FastEthernet0/15 unassigned      YES manual  down    down
FastEthernet0/16 unassigned      YES manual  down    down
FastEthernet0/17 unassigned      YES manual  down    down
FastEthernet0/18 unassigned      YES manual  down    down
FastEthernet0/19 unassigned      YES manual  down    down
FastEthernet0/20 unassigned      YES manual  down    down
FastEthernet0/21 unassigned      YES manual  down    down
FastEthernet0/22 unassigned      YES manual  down    down
FastEthernet0/23 unassigned      YES manual  up      up
FastEthernet0/24 unassigned      YES manual  up      up
GigabitEthernet0/1 unassigned      YES manual  down    down
GigabitEthernet0/2 unassigned      YES manual  down    down
Vlan1          192.168.10.11  YES manual  up      up
```

The output shows that the Port-channel1 is up and operational, and all other interfaces are down. The IP address 192.168.10.11 is assigned to Vlan1.

Verificación en SW2

```
show etherchannel summary
show ip interface brief
```



The screenshot shows a network switch CLI window titled 'SW2'. The 'CLI' tab is selected. The output of the commands is as follows:

```
SW2>
SW2>show etherchannel summary
Flags: D - down          P - in port-channel
       I - stand-alone   S - suspended
       H - Hot-standby (LACP only)
       R - Layer3        S - Layer2
       U - in use        f - failed to allocate aggregator
       u - unsuitable for bundling
       w - waiting to be aggregated
       d - default port

Number of channel-groups in use: 1
Number of aggregators:          1

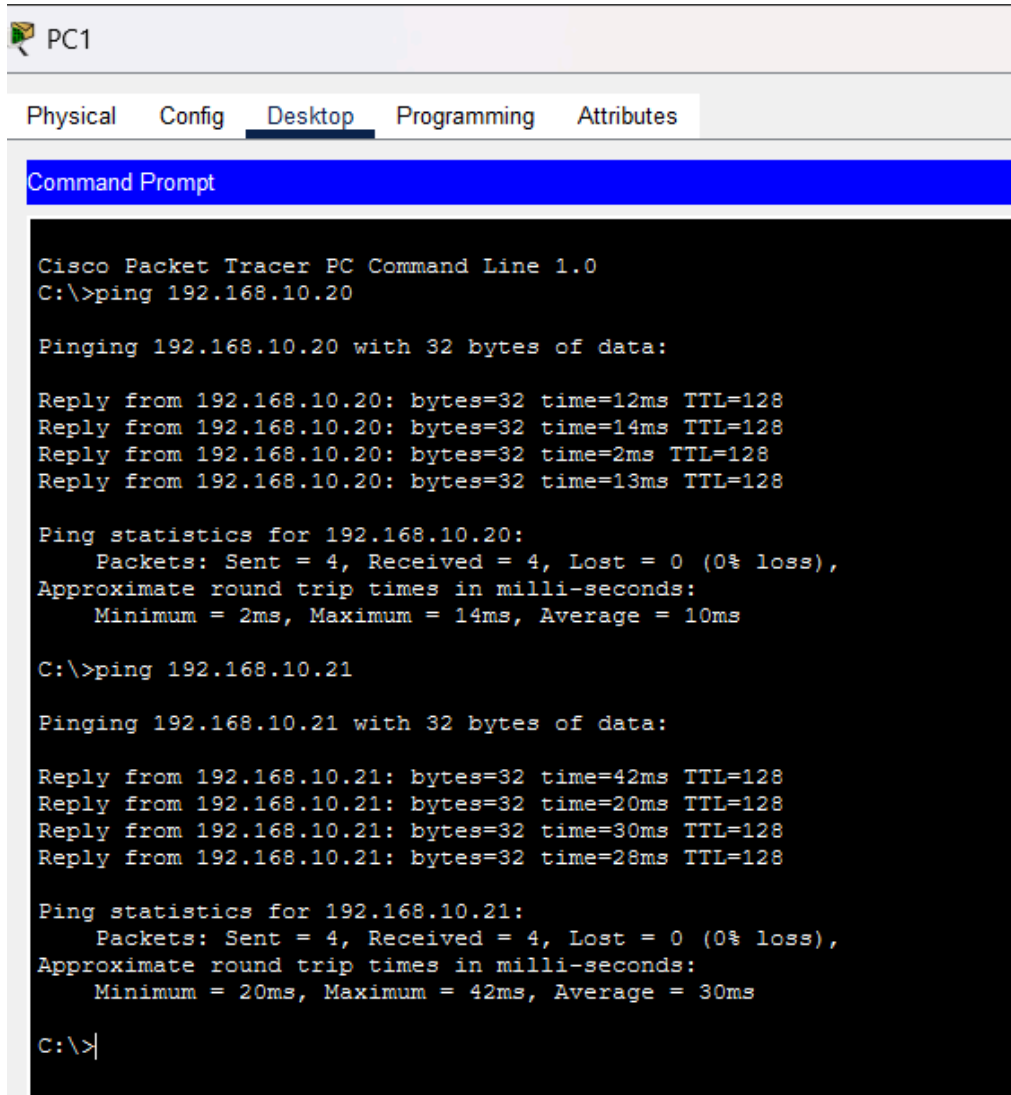
Group  Port-channel  Protocol    Ports
-----+-----+-----+-----
1      Po1(SU)        LACP        Fa0/23(P) Fa0/24(P)
SW2>show ip interface brief
Interface                IP-Address      OK? Method Status  Protocol
Port-channel1            unassigned      YES manual up      up
FastEthernet0/1          unassigned      YES manual down   down
FastEthernet0/2          unassigned      YES manual up      up
FastEthernet0/3          unassigned      YES manual down   down
FastEthernet0/4          unassigned      YES manual down   down
FastEthernet0/5          unassigned      YES manual down   down
FastEthernet0/6          unassigned      YES manual down   down
FastEthernet0/7          unassigned      YES manual down   down
FastEthernet0/8          unassigned      YES manual down   down
FastEthernet0/9          unassigned      YES manual down   down
FastEthernet0/10         unassigned      YES manual down   down
FastEthernet0/11         unassigned      YES manual down   down
FastEthernet0/12         unassigned      YES manual down   down
FastEthernet0/13         unassigned      YES manual down   down
FastEthernet0/14         unassigned      YES manual down   down
FastEthernet0/15         unassigned      YES manual down   down
FastEthernet0/16         unassigned      YES manual down   down
FastEthernet0/17         unassigned      YES manual down   down
FastEthernet0/18         unassigned      YES manual down   down
FastEthernet0/19         unassigned      YES manual down   down
FastEthernet0/20         unassigned      YES manual down   down
FastEthernet0/21         unassigned      YES manual down   down
FastEthernet0/22         unassigned      YES manual down   down
FastEthernet0/23         unassigned      YES manual up      up
FastEthernet0/24         unassigned      YES manual up      up
GigabitEthernet0/1       unassigned      YES manual down   down
GigabitEthernet0/2       unassigned      YES manual down   down
Vlan1                    192.168.10.12  YES manual up      up
SW2>
```

En show etherchannel summary se ve lo siguiente:

- **Po1(SU):** la S indica capa 2 y la U que está en uso.
- **Fa0/23(P) y Fa0/24(P):** la P indica que ambos puertos están agrupados en el port-channel.
- **Protocolo:** LACP.

Prueba ping de PC1 a LAPTOP1 y SMARTPHONE1

```
ping 192.168.10.20  
ping 192.168.10.21
```



The screenshot shows the 'PC1' configuration window in Cisco Packet Tracer, with the 'Desktop' tab selected. A 'Command Prompt' window is open, displaying the results of two ping commands. The first command is 'ping 192.168.10.20', which shows four successful replies with varying times (12ms, 14ms, 2ms, 13ms) and a TTL of 128. The statistics for this ping show 4 packets sent, 4 received, and 0 lost, with an average round trip time of 10ms. The second command is 'ping 192.168.10.21', which also shows four successful replies with times (42ms, 20ms, 30ms, 28ms) and a TTL of 128. The statistics for this ping show 4 packets sent, 4 received, and 0 lost, with an average round trip time of 30ms. The prompt ends with 'C:\>|'.

```
PC1  
Physical Config Desktop Programming Attributes  
Command Prompt  
Cisco Packet Tracer PC Command Line 1.0  
C:\>ping 192.168.10.20  
  
Pinging 192.168.10.20 with 32 bytes of data:  
  
Reply from 192.168.10.20: bytes=32 time=12ms TTL=128  
Reply from 192.168.10.20: bytes=32 time=14ms TTL=128  
Reply from 192.168.10.20: bytes=32 time=2ms TTL=128  
Reply from 192.168.10.20: bytes=32 time=13ms TTL=128  
  
Ping statistics for 192.168.10.20:  
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),  
Approximate round trip times in milli-seconds:  
    Minimum = 2ms, Maximum = 14ms, Average = 10ms  
  
C:\>ping 192.168.10.21  
  
Pinging 192.168.10.21 with 32 bytes of data:  
  
Reply from 192.168.10.21: bytes=32 time=42ms TTL=128  
Reply from 192.168.10.21: bytes=32 time=20ms TTL=128  
Reply from 192.168.10.21: bytes=32 time=30ms TTL=128  
Reply from 192.168.10.21: bytes=32 time=28ms TTL=128  
  
Ping statistics for 192.168.10.21:  
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),  
Approximate round trip times in milli-seconds:  
    Minimum = 20ms, Maximum = 42ms, Average = 30ms  
  
C:\>|
```

Routers y HSRP (CLI)

R1 (activo)

```
enable
configure terminal
hostname R1
interface gigabitEthernet 0/0
  ip address 192.168.10.2 255.255.255.0
  standby 1 ip 192.168.10.1
  standby 1 priority 110
  standby 1 preempt
  no shutdown
exit
end
copy running-config startup-config
write memory
exit
```

R2 (standby)

```
enable
configure terminal
hostname R2
interface gigabitEthernet 0/0
  ip address 192.168.10.3 255.255.255.0
  standby 1 ip 192.168.10.1
  standby 1 priority 100
  no shutdown
exit
end
copy running-config startup-config
write memory
exit
```

Verificación en R1

```
show standby brief
show standby
show ip interface brief
```

R1

Physical

Config

CLI

Attributes

IOS Command Line Interface

```

R1>enable
R1#show standby brief
                P indicates configured to preempt.
                |
Interface  Grp  Pri  P State    Active        Standby        Virtual IP
Gig0/0     1    110 P Active   local         192.168.10.3   192.168.10.1
R1#show standby
GigabitEthernet0/0 - Group 1
  State is Active
    7 state changes, last state change 00:09:21
  Virtual IP address is 192.168.10.1
  Active virtual MAC address is 0000.0C07.AC01
  Local virtual MAC address is 0000.0C07.AC01 (v1 default)
  Hello time 3 sec, hold time 10 sec
  Next hello sent in 1.457 secs
  Preemption enabled
  Active router is local
  Standby router is 192.168.10.3
  Priority 110 (configured 110)
  Group name is hsrp-Gig0/0-1 (default)
R1#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0  192.168.10.2    YES manual up          up
GigabitEthernet0/1  unassigned      YES unset  administratively down down
GigabitEthernet0/2  unassigned      YES unset  administratively down down
Vlan1          unassigned      YES unset  administratively down down
R1#

```

Verificación en R2

```

show standby brief
show standby
show ip interface brief

```

R2

Physical

Config

CLI

Attributes

IOS Command Line Interface

```

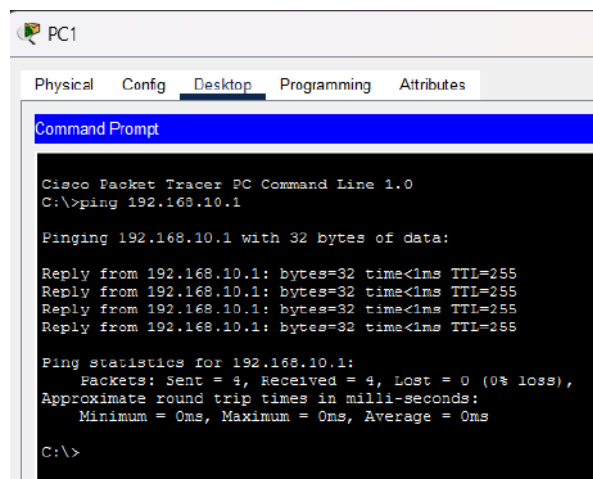
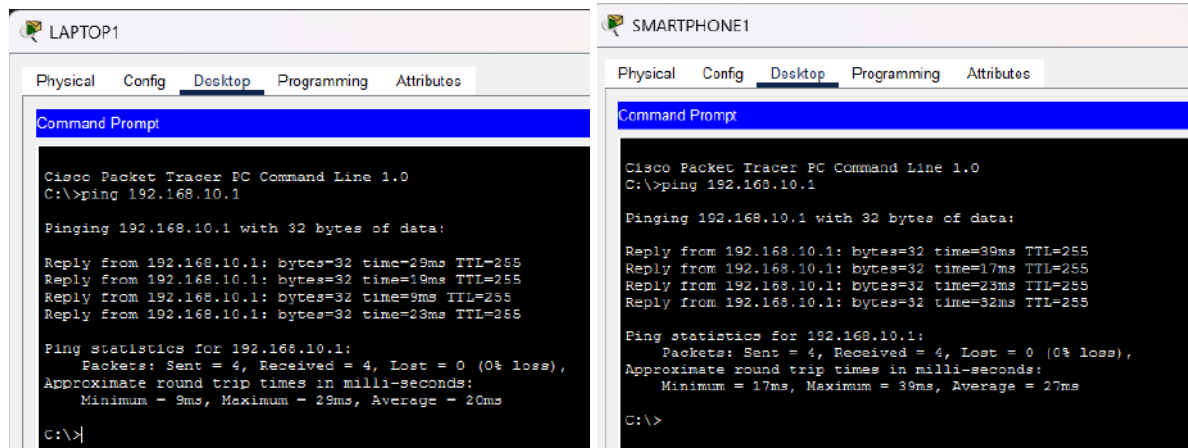
R2>enable
R2#show standby brief
                P indicates configured to preempt.
                |
Interface  Grp  Pri  P State    Active        Standby        Virtual IP
Gig0/0     1    100 Standby  192.168.10.2  local         192.168.10.1
R2#show standby
GigabitEthernet0/0 - Group 1
  State is Standby
    7 state changes, last state change 00:10:01
  Virtual IP address is 192.168.10.1
  Active virtual MAC address is 0000.0C07.AC01
  Local virtual MAC address is 0000.0C07.AC01 (v1 default)
  Hello time 3 sec, hold time 10 sec
  Next hello sent in 2.561 secs
  Preemption disabled
  Active router is 192.168.10.2
  Standby router is local
  Priority 100 (default 100)
  Group name is hsrp-Gig0/0-1 (default)
R2#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0  192.168.10.3    YES manual up          up
GigabitEthernet0/1  unassigned      YES unset  administratively down down
GigabitEthernet0/2  unassigned      YES unset  administratively down down
Vlan1          unassigned      YES unset  administratively down down
R2#

```

Pruebas de conectividad

Desde PC1, LAPTOP1 y SMARTPHONE1, haz ping a la IP virtual:

```
ping 192.168.10.1
```



Conclusiones

1. La redundancia de gateway con HSRP garantiza la continuidad del servicio. Al configurar R1 y R2 en el grupo HSRP 1 con una IP virtual (192.168.10.1), los hosts no dependen de un solo router. Cuando el router activo falla, el standby deja de recibir mensajes hello, asume el rol activo y los clientes siguen usando el mismo gateway sin cambiar su configuración. La prioridad y el preempt permiten además controlar qué router debe ser el activo y recuperar ese rol cuando vuelve a estar disponible, lo que da un comportamiento predecible.
2. EtherChannel con LACP aumenta la disponibilidad y el ancho de banda del enlace entre switches. Agrupar Fa0/23 y Fa0/24 en el Port-channel 1 hace que ambos cables funcionen como un único enlace lógico. Esto duplica la capacidad disponible y evita que la caída de un cable interrumpa la comunicación, porque el tráfico continúa por el enlace restante. Además, LACP negocia la agrupación entre ambos extremos, lo que reduce el riesgo de errores de configuración frente a un enlace estático.
3. La alta disponibilidad se logra combinando redundancia en varias capas, y se debe verificar con pruebas. En esta práctica, EtherChannel protege la capa 2 (enlaces entre switches) y HSRP protege la capa 3 (gateway), mientras que el AP con WPA2 permite integrar clientes inalámbricos de forma segura a la misma red. Simular fallos apagando interfaces y comprobar con comandos como `show standby brief` y `show etherchannel summary` demuestra que la red mantiene la conectividad. Sin esas pruebas, una configuración redundante podría parecer correcta y fallar cuando realmente se necesita.